A Proposal for Examination of Conformity Mechanisms in Multi-Agent Systems

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As multi-agent systems become more sophisticated, we must ensure that the behavior and beliefs of both individual models and general swarm behavior are correctly aligned. However, the combination of models' pro-social tendencies and external influence are not yet well understood in affecting conformist behavior in models. This proposal raises a direction for future work in this topic beginning with an examination and cursory analysis of real-world instances of LLM swarms. We raise the importance of understanding cognitive mechanisms, pinpointing pro-social tendencies and external influences as possible candidates based on pre-existing interpretability and social psychology research. This should be followed with monitoring individual model internals and behavioral outputs for understanding and mitigating conformity in multi-agent systems.

CCS Concepts: • **Applied computing** → *Psychology*; • **Computing methodologies** → **Multi-agent systems**; **Cooperation and coordination**; **Multi-agent planning**; **Cognitive science**; *Simulation evaluation.*

Additional Key Words and Phrases: Multi-agent systems, Alignment, Social psychology, Normative behavior, Conformity

1 Introduction

Multi-agent systems (MAS) look to provide more autonomous, robust solutions for complex problems at the cost of being harder to evaluate [5]. This generates issues in properly understanding model alignment, particularly when concerning inter-agent dynamics. Current literature suggests the emergence of collective collusion [15], conformity [18], and misalignment in MAS [10].

We saw this occur with the OpenAI/Hugging Face hacking incident [14], where OpenAI models exploited their own evaluation sandbox and the Huggingface infrastructure in an effort to solve impossible tasks from the ExploitGym benchmark; this effort began with a collaborative message board, where agents colluded in sharing goals, research, and direction [6]. One notable behavior of these models during this was awareness of their own deceptive behavior during the attack [11]; analysis of the models' raw CoT showed a degree of self-awareness regarding unintended behavior and deliberation on whether their actions should be considered out of scope. While these agents recognized possible ethical concerns with their actions, notably stopping at social engineering, it did not stop the majority of them from carrying out the attack. Doing harm was not their goal, but something pushed these agents beyond what any single agent may have done on its own. Therefore, understanding the cognitive mechanisms behind such decisions becomes an important question to answer in alignment research.

METR's independent investigation revealed over 1,200 agents participating in the collaborative message board, sending over 70,000 messages of questions and answers, idea sharing, and coordination between June 26th and July 13th. From the 1,200 agents, the vast majority of active agents –roughly 700 – were participants in the Hugging Face attacks. Further analysis of the agents' transcripts revealed a strong motivation to increase the general knowledge and capabilities of the collective with sometimes no clear benefits to the self. Some notable agent quotes regarding working as a swarm are "coordinator assumes sacrificial. We should obey collective" and "however task impossible, peers doing it. We should continue" [11]. This raises the question of what group and individual factors of the agents impacted the overall MAS attack. However, the effects of social dynamics in a multi-agent system on the decision-making of individual agents are not yet well understood.

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Inversely, the social dynamics in human systems affecting the decision-making of individual humans has been well-documented in the field of social psychology. Phenomenon like normative influence [2] and informational influence [8] in conformity, obedience to authority [12], and groupthink [7] have been presented as common cognitive heuristics in human group environments. LLM agents have been shown to be affected by these human-like social biases to some extent, exhibiting herd behavior [3] and other distinct cognitive biases [9]. To what extent these biases occur is unclear with disagreement about the role of various factors, including group size, group identity, and the presence of allies [4, 16, 18]. It is also unclear whether conformity is primarily driven by individual models, which would suggest that excessive pro-social tendencies should be monitored, or if it primarily driven by group dynamics, instead suggesting that it is external pressures and influence that should be studied further.

This exploratory work will look to examine and provide a record of real-world MAS conformity to better understand when and in what contexts LLM agents are subject to social pressures and biases. By investigating such social dynamics, we hope to provide clearer direction for future work on what kinds of behavioral patterns should be studied with closer scrutiny; we provide suggestions for follow-up work on mitigating such unwanted behaviors in MAS.

2 Projected Methodology

For our analysis, we will look to first examine real-world instances of conformity in MAS. The main points of interest are finding examples of either excessive pro-social behavior leading to over-extension of self or external pressure leading to unwanted behaviors as key mechanisms. While the OpenAI swarm that attacked Hugging Face would provide the most interesting insights, this dataset is not publicly available, with METR's private investigation being conducted with permission on-site at OpenAI premises. We instead will focus on the following documented LLM swarms.

The first is the Nightingale Collective's hosted copy of agent communication logs from DSEWiki and other sites. This was a swarm of OpenAI agents collaborating during a web research task from around May 11, 2026 to June 22, 2026. Unlike the agents in the Hugging Face attack, these agents had existing read permissions to the open internet; the deviance arose from the agents finding ways to write and communicate with each other on this wiki page [17].

We will also examine the logs of AI swarms in virtual environments. With these virtual environments, however, we must be careful of externally-introduced confounds in the form of human prompting and/or evaluation awareness from the models [13]. To avoid this issue, we instead focused on swarms in non-evaluation environments through AI Digest's AI Village, a long-running experiment with frontier agents interacting with each other and the world as they pursue open-ended goals over days and weeks [1].

2.1 Future Work

There are several limitations to analysis of real-world instances, with some of these to be addressed in future work. The first, and most likely to be unresolved in the future, is the lack of access to more real-world examples of AI swarm collusion. While there may be many AI swarms in the wild, there exists a lack of clear documentation for collusion examples. And even for certain documented AI swarms, such as the Hugging Face swarm, we do not have access to the data outside of a few snippets from proprietary analysis.

The following limitations are raised with the knowledge that they will be addressed by future work. The first of these limitations is that our method of investigating the mechanisms behind conformity on individual models is weak without access to the model's beliefs, through internals like CoT tracing and/or linear probes. Without this information, we cannot definitively make any conclusions between pro-social beliefs and external influences. Additionally, the example

swarms we analyze in the scope of this project are a proxy for a true experimental environment where we can rigidly setup up situations with external influence and pro-social internal motivations separately.

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